Linear Interpolation Formula:
For two points on (i,PW): $(i_1,PW_1) = (w,x)$ and $(i_2,PW_2) = (y,z)$, the formula for $(i^*,0)$ is:
$(i^*,0) = (i^*=y-(y-w)*(0-z)/(x-z), 0)$

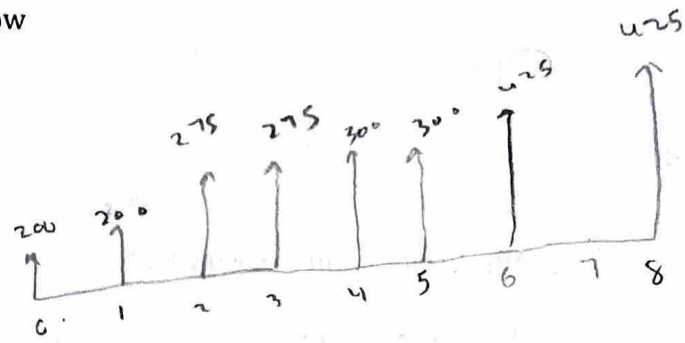
Quadratic Formula:
The solution to the quadratic equation $aX^2 + bX + c = 0$ is $X = [-b +/- (b^2-4ac)^{1/2}]/2a$

The Chapter 4 formulas provided are:
$I_k=r/M$ and $i_k=(1+r/M)^c-1$ and $c*k=M$
(it's your responsibility to know what i,r,M,c and k stand for!)

3) Consider the following cash flow: \$200 at t=0 and 1, \$275 at t=2 and 3, \$300 at t=4 and 5, and \$425 at t=6 and 8 (nothing at t=7). Now consider a second cash flow of \$X at t=0,1,2,3,4,5 and 6. Find the unknown value of X that satisfies the equivalence.

9 %

a) None of the below
b) 287-297
c) 297-307
d) 307-317
e) 317-327



X

$$P = 200 + \frac{200}{(1+9\%)^1} + \frac{275}{(1+9\%)^2} + \frac{275}{(1+9\%)^3} + \frac{300}{(1+9\%)^4} + \frac{300}{(1+9\%)^5} + \frac{425}{(1+9\%)^6} + \frac{425}{(1+9\%)^8}$$

1234.8

+

9 1701.5

$$V + X\,(P.A.,\ 9\%,\ 5) = 1701.5$$

347.475

4) You decide to save $2000 a year for retirement starting on your 46th birthday. Your last (20th) deposit is on your 65th birthday. On that day you decide to withdraw $X and you continue to withdraw each year, increasing the amount by $1000 per year for 10 years (so 11 withdrawals in all). Find X.

a) 4605-4620
b) 4620-4635
c) 4635-4650
d) 4650-4665
e) None of the above

$2000(F/A, 9\%, 20) = X + X(P/A, 9\%, 10) + 1000(P/G, 9\%, 10)(1+9\%)$

$2000(51.1601) = X + X(6.4177)$

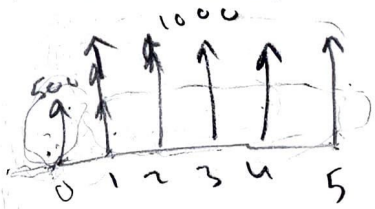
$2000(1+9\%)^{20} + 2000(F/A, 9\%, 19) = [X(P/A, 9\%, 10) + X] + [1000(P/G, 9\%, 11)](1+9\%)$

$103245.8215 = 6.4177X + X + [30796.429]$

$X = 9767.9$

5) Consider the following cash flow: t=0: $500, t=1-5: $1000
In computing F at the end of year 5 at an interest rate of 9%, which of the following equations is ***incorrect***?

a) F= [$1000(P/A,9%,6) – 500/1.09)](F/P,9%,6)
b) F=[$500 + $1000(P/A,9%,5)](F/P,9%,5)
c) F=[$500(A/P,9%,5) + $1000](F/A,9%,5)
d) F=$500(F/A,9%,5) + $500(P/A,9%,5)(F/P,9%,5)
e) All the equations are correct



Chapter 4

6) The bank offers you the following borrowing rate: '4% compounded monthly'. What is the effective monthly rate?

a) 0.25%
b) 0.3333%
c) 0.7667%
d) 2.5%
e) None of the above

7) Continuing from Q#6, what is the effective weekly rate?

a) 0.0007282
b) 0.0007382
c) 0.0007482
d) 0.0007582
e) None of the above

$\left(1+\frac{4\%}{12}\right)^{\frac{1}{4}}-1$

8) The interest rate is 9% compounded monthly. You borrow $15,000 to be paid off in equal monthly installments over 5 years. What is the payment?

a) 311-313
b) 313-315
c) 315-317
d) 317-319
e) None of the above

$15000 = X\,(PIA, \overset{0.75}{\frac{9}{12}}, 60)$

$15000 = X * 48.1734$

$X = 311.37$

~~4.75%~~

9) The interest rate is 9% compounded monthly. You borrow $15,000 to be paid off in equal quarterly installments over 3 years. What is the payment?

a) 1440-1444
b) 1444-1448
c) 1448-1452
d) 1452-1456
e) None of the above

$i_q = \left(1 + \frac{9\%}{12}\right)^3 - 1$
$= 0.022669$

$n = 3 \times 4 = 12$

$15{,}000 = x\,(P|A, i_q, 12)$

$x = 1441.7$

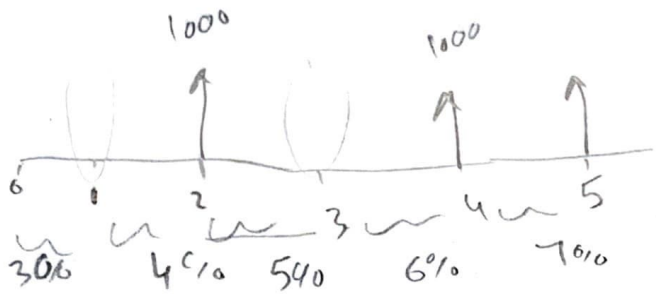
10) Continuing from Q#9: At the end of the 1st year (after 4 payments) you want to pay off the loan in its entirety. What is the balance still owed on the loan?

a) 9998-10,003
b) 10,139-10,143
c) 10,298-10,303
d) 10,339-10,343
e) None of the above

$B = A(P|A, i, N-n)$
$= 1441.74721 \times (P|A, i, 12-4)$
$= 10{,}441.03$

11) The interest rate in the first year is 3% (between t=0 and t=1), the second year is 4%, and continuing to rise by 1% per year thereafter. You deposit $1000 in the account at t=2,4 and 5. What is the equivalent equal sum payment of $X at t=1,3 and 4 that makes the two cash flows equivalent?

a) None of the below
b) 945-949
c) 949-953
d) 953-957
e) 957-961



2260.91

$1000(1+5\%)(1+6\%)(1+7\%) + 1000(1+7\%)$

=

$X(1+4\%)(1+5\%)(1+6\%)(1+7\%) + X(1+6\%)(1+7\%)$

1.2385464 + 1.1342

$952.86

12) A new bond is issued with a $10000 face value, a 6% annual coupon payment and a 20 year maturity date. You purchase the bond (you are the original buyer) for its face value. Immediately after the 5th payment you wish to sell the bond but the interest rate in the economy has changed to 9% (this means new bonds will be issued with 9% coupon payments). What can you sell your bond for?

a) None of the below
b) $7557-7567
c) $7567-7577
d) $7577-7587
e) $7587-7597

13) Continuing from Q#12: Suppose the new buyer holds the bond until its maturity date (collecting all of her payments during that time). What will be the new buyer's rate of return (you can think of this as an IRR)?

a) None of the below
b) 6.75%-7.75%
c) 7.75%-8.75%
d) 8.75%-9.75% ✓
e) 9.75%-10.75%

800

9%

0 = -7581.8 + 600 (P|A, 8%, 15) + 10,000 $(1+i)^{-15}$

8.0607

900

= $7581.8

8

8.75

7750

7581

6957

6

7

9

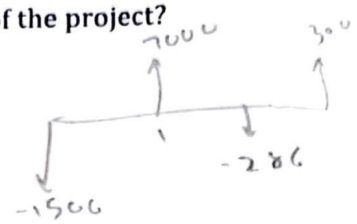
-1954

9%

10

Chapter 5

14) The MARR is 10%. Consider a project with the following cash flow: t=0: -1500, t=1: +2000, t=2: -286. What is the IRR of the project?



a) 13-15%
b) 15-17%
c) 17-19%
d) 19-21%
e) None of the above

-1500

$((-1500(1+i\%) + 2000)(1+0.1) - 286)(1+i) + 300 = 0$

$\boxed{16}$

18

19 = -33 -44

15) Continuing from Q#14: Add another cash flow: t=~~4~~3: +300. Now what is the IRR of the project?

a) None of the below
b) 21-23%
c) 23-25%
d) 25-27%
e) 27-29%

22% = 179.22

29% = 23.295

30%

Bonus: 16) What is the seller's IRR for the bond described in Q#12 supposing that he sold the bond for $10,000 to the new buyer (this is not the answer for Q#12)?

a) 6.5-7.5%
b) 7.5-8.5%
c) 8.5-9.5%
d) 9.5-10.5%
e) None of the above

6.5 9529

7 90829

$\boxed{6\%}$